

Predicting Airline Customer Satisfaction

- Logistic Regression Insights from 25,976 Records
- Presented by: Jeremy Juke
- Course: BIA650
- Date: 06/29/2025



Business Problem & Hypothesis



- DECLINE IN CUSTOMER SATISFACTION AND RETENTION, ESPECIALLY IN LOWER-TIER CABINS

- ALTERNATIVE HYPOTHESIS: INCREASED FOOD AND DRINK OPTIONS WILL POSITIVELY INFLUENCE CUSTOMER SATISFACTION AND RETENTION.

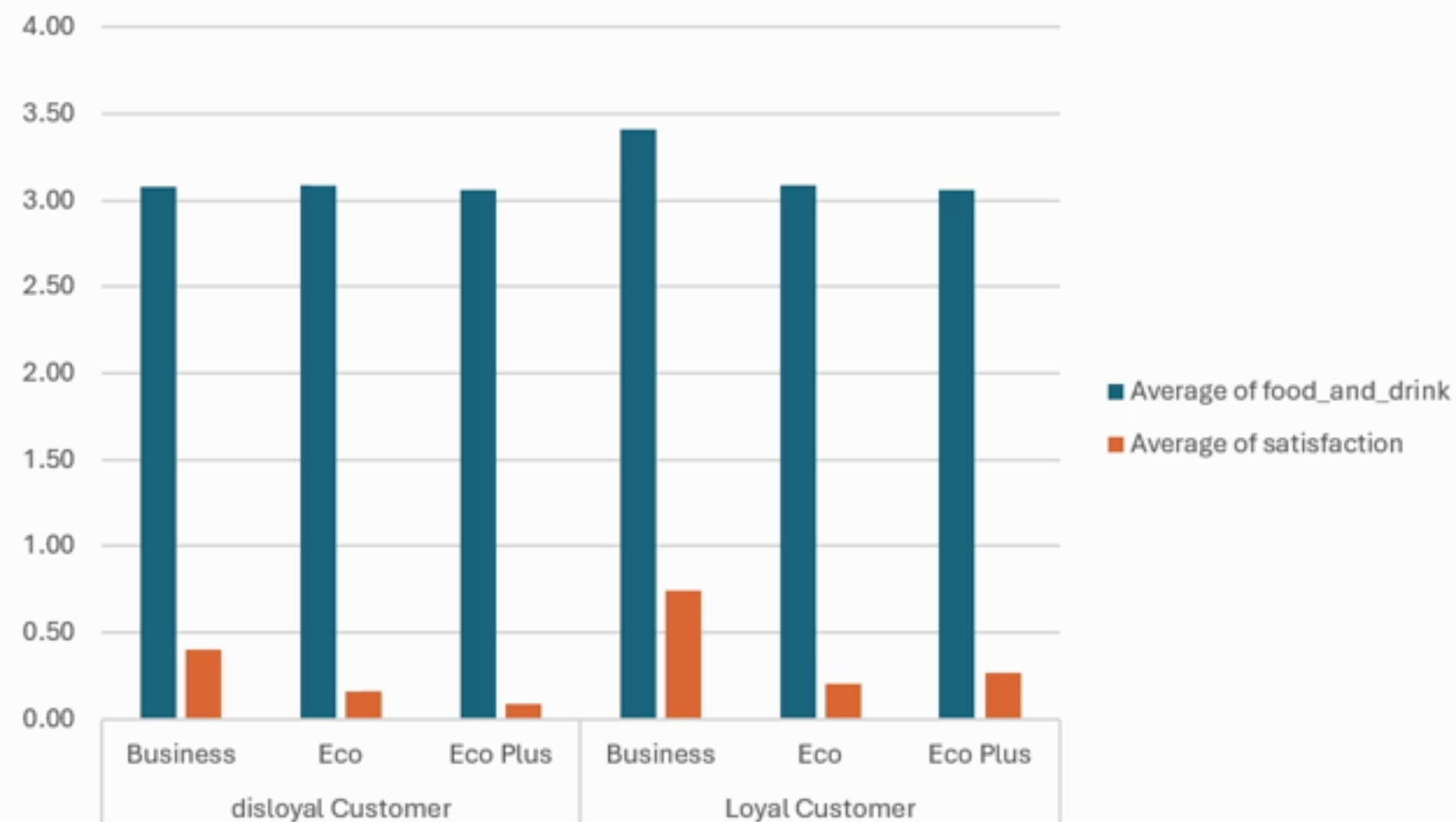
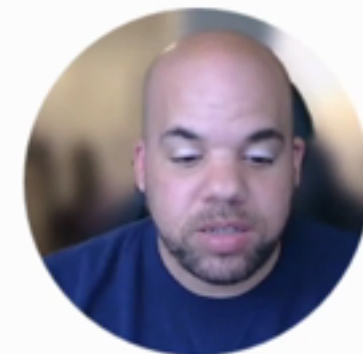
- NULL HYPOTHESIS: FOOD AND DRINK DO NOT INFLUENCE SATISFACTION OR RETENTION

GOAL: USE PREDICTIVE MODELING TO VALIDATE THIS HYPOTHESIS

Avg. Food & Satisfaction Scores by Class and Loyalty

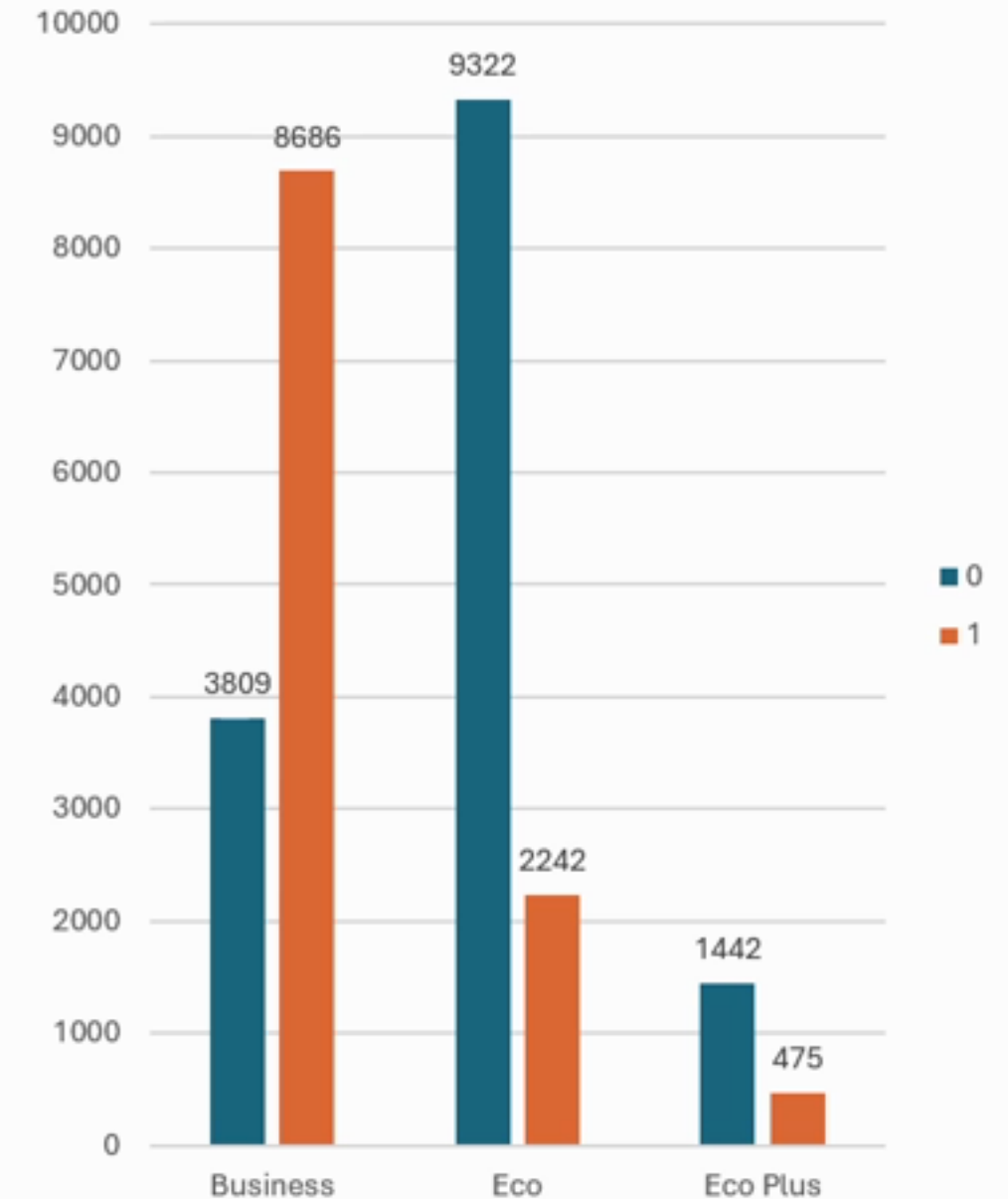
Key Takeaways:

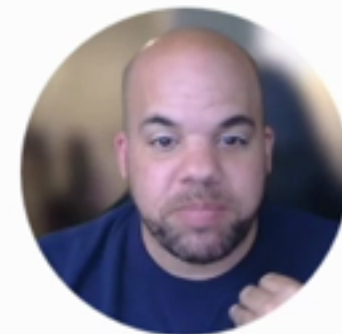
- Loyal business class passengers have the highest satisfaction
- Disloyal customers show flat ratings regardless of class for Food Scores.
- Eco has the lowest satisfaction with loyal customers and eco plus has the lowest satisfaction with disloyal customers.



Satisfaction Count by Class (Satisfied vs Not)

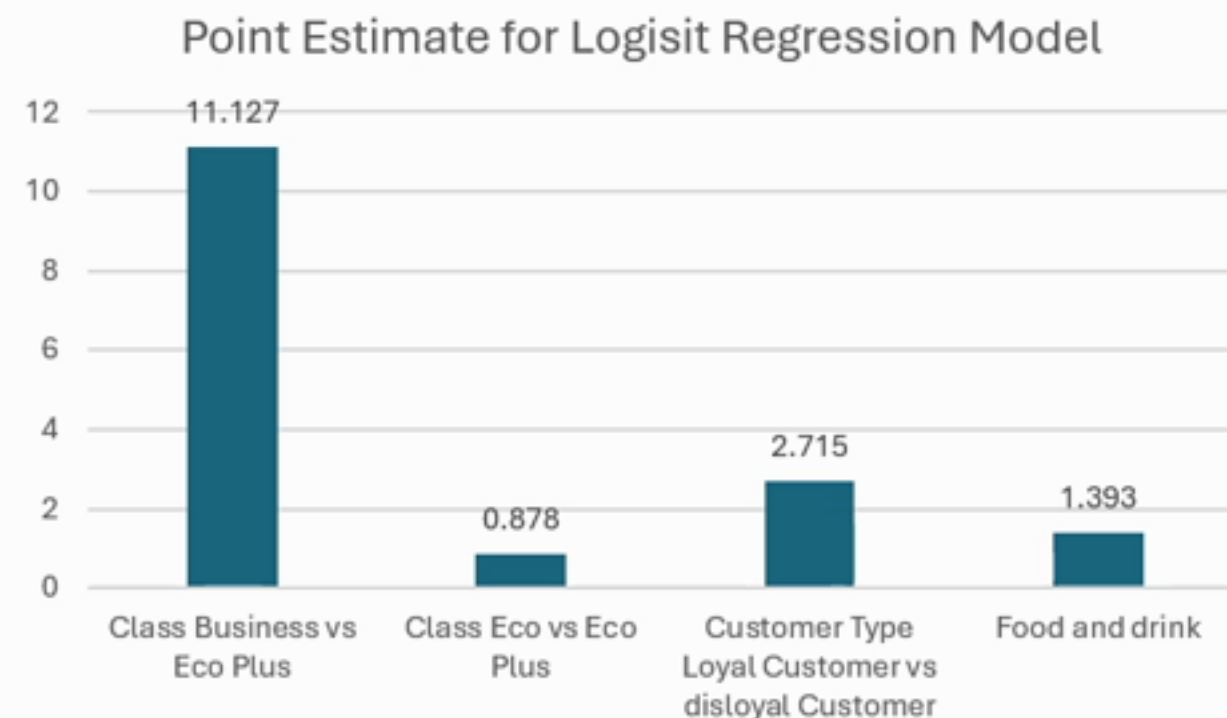
- Clarify:
- - 0 = Dissatisfied
- - 1 = Satisfied
- Emphasize:
- - Business class has the highest satisfaction representing 33% of the total data population.
- - Eco class were the highest dissatisfied representing 35% of the total data population.





Point Estimates from Logistic Regression Model

- Key findings:
- - Business class = 11x satisfaction odds vs Eco Plus
- - Food & drink score = 39% increase per unit
- - Model concordance = 81.3% (strong predictive quality)



Association of Predicted Probabilities and Observed Responses

Percent Concordant	81.3	Somers' D	0.667
Percent Discordant	14.6	Gamma	0.696
Percent Tied	4.1	Tau-a	0.329
Pairs	7.0328275E17	c	0.833

Next Steps for Implementation



1

- Pilot food enhancements in Eco and Eco Plus cabins.

2

- Collect post-implementation data through follow-up sentiment surveys.

3

- Retrain the model with updated data and validate probability estimates for eco and eco plus.

Thank You



QUESTIONS?



LET'S DISCUSS NEXT
STEPS.